

Experiment No: 03
Experiment Name: Verification of Kirchhoff's Voltage law and Voltage divider rule.
Course Code: EEE-1209
Course Title: Electrical Circuit lab

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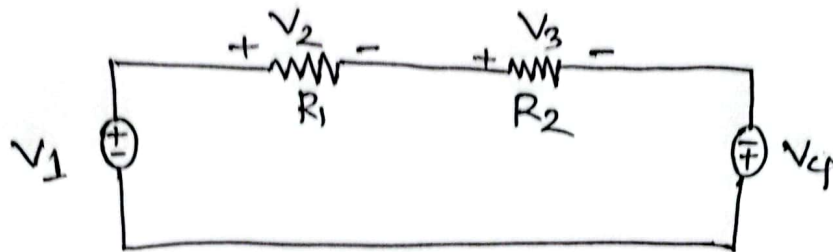


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Experiment name: Verification of Kirchhoff's voltage Law and voltage divider rule.

KVL theory:

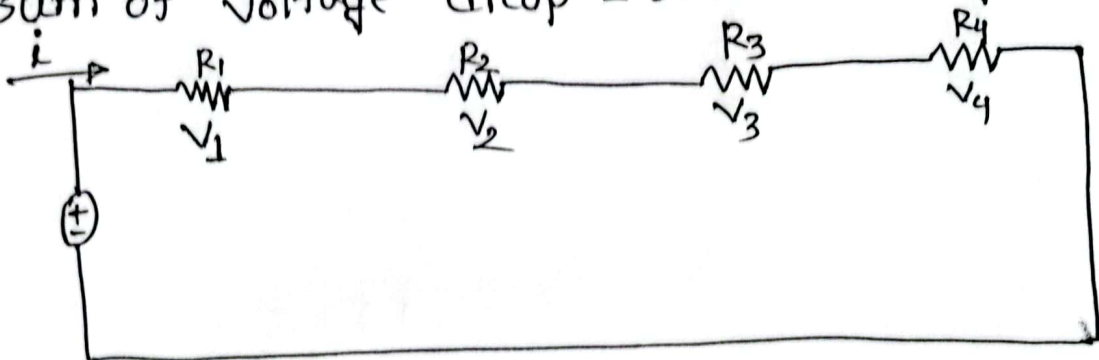
In a closed electric circuit the algebraic sum of potential drops is equal to the algebraic sum of total electromotive force occurring round the circuit.



Here,

$$-V_1 + V_2 + V_3 - V_4 = 0$$
$$V_1 + V_4 = V_2 + V_3$$

sum of voltage drop = sum of voltage rise.



$$V_1 = \frac{R_1}{R_1 + R_2 + R_3 + R_4} \times V$$

$$V_2 = \frac{R_2}{R_1 + R_2 + R_3 + R_4} \times V$$

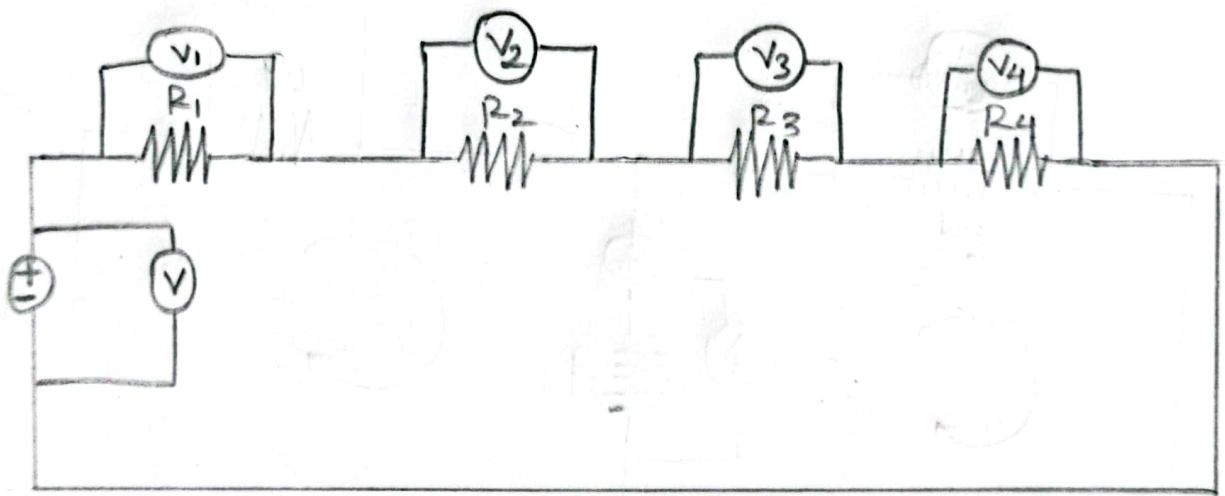
According to this,

$$V_n = \frac{R_n}{R_1 + R_2 + \dots + R_n}$$

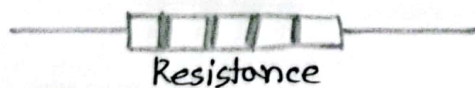
Required Equipments:

- (1) Resistor
- (2) Ammeter
- (3) Voltmeter
- (4) Bread board
- (5) DC power supply
- (6) Connecting wires

Circuit Diagram:



Verification of KVL.



Procedure:

- (1) Connect the circuit in the trainer kit as per circuit diagram.
- (2) Adjust the input voltage by adjuster

for different reading. (8V, 10V & 12V)

(3) Connect the voltmeters to get the required voltage.

(4) Repeat the same procedure for different observation.

(5) Compare the value with theoretical results.

Calculation:



According to the colours,

$$R_1 = 1k\Omega$$

$$R_2 = 10k\Omega$$

$$R_3 = 33k\Omega$$

$$R_4 = 4.7k\Omega$$

Result and Discussion:

For,

$$V = 8,$$

$$V_{1c} = \frac{R_1}{R_1 + R_2 + R_3 + R_4} \times V_s = \frac{1}{1 + 10 + 33 + 4.7} \times 8 = 0.1643$$

$$V_{2c} = \frac{R_2}{R_1 + R_2 + R_3 + R_4} \times V_s = \frac{10}{1 + 10 + 33 + 4.7} = 1.6427$$

$$V_{3c} = \frac{R_3}{R_1 + R_2 + R_3 + R_4} \times V_s = \frac{33 \times 8}{48.7} = 5.421$$

$$V_{4c} = \frac{R_4}{R_1 + R_2 + R_3 + R_4} \times V_s = \frac{4.7 \times 8}{48.7} = 0.772$$

For $V=10$,

$$V_{1c} = \frac{1}{48.7} \times 10 = 0.205$$

$$V_{2c} = \frac{10}{48.7} \times 10 = 2.05338$$

$$V_{3c} = \frac{33}{48.7} \times 10 = 6.78$$

$$V_{4c} = \frac{4.7 \times 10}{48.7} = 0.965$$

For $V=12$,

$$V_{1c} = 0.2464$$

$$V_{2c} = 2.464$$

$$V_{3c} = 8.1314$$

$$V_{4c} = 1.16$$

Data Table:

No. of observation	V_s	V_{1c}	V_{2c}	V_{3c}	V_{4c}	V_{1m}	V_{2m}	V_{3m}	V_{4m}	$V_m = V_{1m} + V_{2m} + V_{3m} + V_{4m}$	$\frac{V_s - V_m}{V_s} \times 100$
1	8	0.16 45	1.6 25	5.42 1	0.77 2	0.16 5	1.6 25	5.44	0.7 66	7.996	0.05%
2	10	0.20 5	2.05 338	6.78	0.96 5	0.20 5	2.02	6.76	0.9 52	9.95	0.15%
3	12	0.24 69	2.46 4	8.13 14	1.16	0.24 6	2.42 4	8.11	1.14 3	11.94	0.15%

Here,

V_s is simulated to V_m . So for verification of KVL and voltage divider rule we made a series circuit with two four circuit. Now measure current with ammeter. Then we measure four voltage across four resistors. Sum of four voltage is equal to the supply voltage. By this equality we can say that the voltage law is verified by our experiment. Now again we put all value in voltage divider equation and also now we get approximately same value. And by getting same value in all types we can said that by this circuit we verify the KVL and voltage divider rule.

Conclusion:

From this result, we can see that the algebraic sum of potential drops is equal to the algebraic sum of total electromotive force. So, we can say that Kirchhoff's voltage law is true in real world.

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